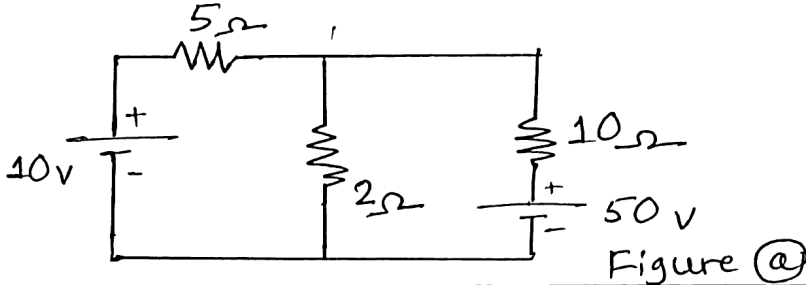
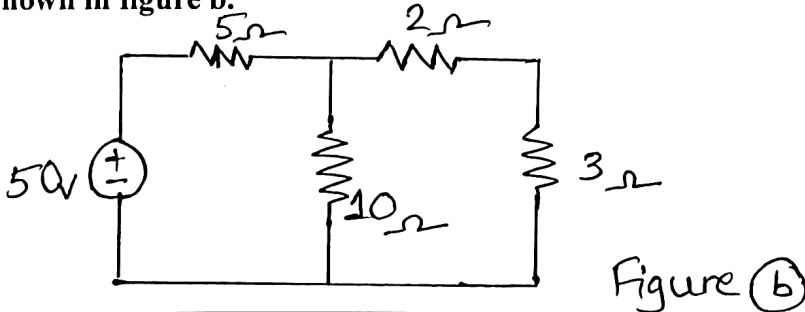
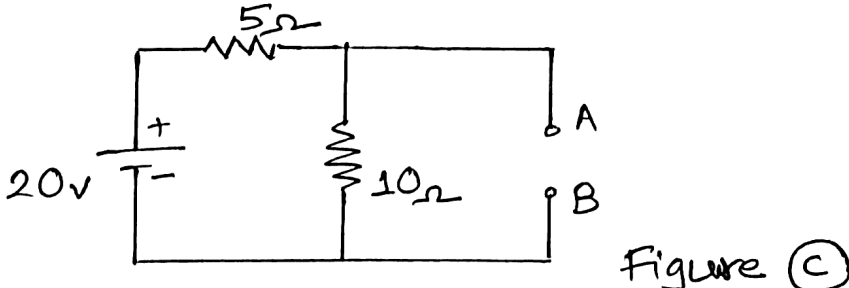


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Electronics Engineering Semester :III Subject Code & Name: Network Theory (BTEXC304) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		CO	Marks
Q.1	Solve Any Two of the following.		
A)	Determine the current flowing through 5Ω resistor in the network shown in figure a.  Figure (a)	CO1	6
B)	Use Thevenin's theorem to find the current flowing through 3Ω resistor in the network shown in figure b.  Figure (b)	CO1	6
C)	Determine Norton's equivalent circuit for the given circuit shown in figure c.  Figure (c)	CO1	6
Q.2	Solve Any Two of the following.		
A)	A series RL circuit with $R=30\Omega$, $L=15H$ and $V=60\text{ v}$ applied at $t=0$, determine current i.	CO2	6
B)	A series RC circuit with $R=50\Omega$, $C=5\mu F$ and $V=200\text{ v}$ applied at $t=0$, determine current i.	CO2	6

C)	For circuit shown in figure d find current equation when switch s is opened at $t=0$.	CO2	6
Figure (d)			
Q. 3	Solve Any One of the following.		
A)	Find Laplace Transform of the given functions a) $L(3 \sin 2t + 4 \cos 3t)$ b) $L(\frac{1}{5} \sin 5t)$	CO3	6
B)	Find Final value for given function a) $f(t)=2 + e^{-3t} \cos 2t$ b) $f(t)=6(1-e^{-t})$	CO3	6
C)	Find initial value for given function a) $f(t)=5e^{-4t}$ b)	CO3	6
Q.4	Solve Any Two of the following.		
A)	Derive the expression of Z Parameters in terms of Y parameters	CO4	6
B)	Find Z parameters for the circuit shown in figure e.	CO4	6
Figure (e)			
C)	For the network shown in figure f, determine the transfer functions $G_{21}(s)$ and $Z_{11}(s)$ and driving point admittance $Y_{11}(s)$.	CO4	6
Figure (f)			
Q. 5	Solve Any One of the following.		
A)	Derive the expression of characteristic impedance(Z_{0T}) of filter T network	CO5	6
B)	Explain different types of Filters	CO5	6
C)	Design LPF (Tsection) having cut-off frequency 2KHz and load resistance 500Ω	CO5	6
*** End ***			